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Contributions of the PCL/-R to the Construct Validation and Mechanistic Understanding of the Cleckley Psychopath

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Abstract

In this article, we detail the contributions of the Psychopathy Checklist (PCL) and the PCL-Revised (PCL/-R; Hare, 2003) for the scientific study of psychopathy and highlight its role in bringing order and clarity to the field. We explore how the introduction of the PCL/-R facilitated the construct validation of and explorations of mechanistic underpinnings of the Cleckley psychopath, focusing specifically on the response modulation hypothesis (RMH; Patterson & Newman, 1993). We argue that the PCL/-R has been critical for advancing research on the psychopath initially outlined by Cleckley (1941) in *The Mask of Sanity*. In discussing current debates surrounding use of this measure, we advocate for the continued use of the PCL/-R as well as the development and use of alternative assessment measures, proposing a method for integrating and advancing different conceptual and measurement approaches. Overall, we call for careful consideration, delineation, and comparison of hypothetical constructs being studied in modern psychopathy research through the provision of evidence for the construct validity of the construct(s) being studied.

Keywords Psychopathy · Cleckley · Construct validation · Assessment

Once deemed the "gold standard" for the assessment of psychopathy (p. 429, Acheson & Payne, 2005), the Psychopathy Checklist (PCL) and its successor (the PCL-Revised; PCL/-R) rose to prominence as a central instrument in both applied forensic and academic research domains. This measure was designed to capture core interpersonal, affective, and behavioral attributes that exemplify the psychopathic individual described by Hervey Cleckley's (1941) *The Mask of Sanity*.[1] While the instrument continues to dominate the forensic literature (e.g., Le et al., 2023; McCarthy et al., 2024; Moosburner et al., 2024; Moretti et al., 2024; Sohn & Lee, 2023), its application to specifying the construct validity of psychopathy as well as its centrality to research on the underlying causal processes have decreased because of alternative approaches to studying psychopathy that emphasize constituent traits (e.g., Jakobwitz & Egan, 2006; Lilienfeld, 2018; Lilienfeld et al., 2015; Lynam & Miller, 2015; Miller et al., 2015; Miller & Lynam, 2015; Patrick et al., 2012; Strickland et al., 2013). The goal of this article is to highlight the contributions of the PCL/-R for the scientific study of psychopathy, review current debates surrounding use of the measure, and facilitate the integration of diverse approaches to psychopathy assessment. We discuss the development of the construct of 'the psychopath,' contrasting the early lack of standardization in definition and assessment with the pivotal formulation by Cleckley. We describe

[1] Hare states he "did not base the items on the simple, uncritical acceptance and application of the 16 characteristics listed in Cleckley's (1976) Clinical Profile of psychopathy. Rather, they were based on a deep appreciation of the rich clinical material he provided (some an interpretation and integration of other clinical writings), 15 years of experience and empirical research by Hare and his colleagues and students, and theoretical and empirical studies by other clinical and researchers" (p. 48, Hare et al., 2018). While Hare integrated additional information and data to update the notion of the Cleckley psychopath, the core features are preserved; this claim is supported by the PCL's high (e.g., 0.83) correlation with Cleckley ratings (see Hare, 1980). As such, the term Cleckley psychopath will be retained.

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how the introduction of the PCL/-R facilitated the construct validation of and explorations of mechanistic underpinnings of the Cleckley psychopath, focusing specifically on the development of the response modulation hypothesis (RMH; Patterson & Newman, 1993) and its derivatives. Next, we introduce recent criticisms of use of the PCL/-R and highlight how they have created a false dilemma in the field. We argue that, by providing a reliable assessment based on the Cleckley psychopath, the PCL/-R has been critical for advancing research on the mechanisms underlying the syndrome and has directly facilitated a better understanding of psychologically based targets for intervention and remediation. We outline how use of different methodologies may result in the measurement and study of divergent and unintended theoretical constructs, thereby highlighting the increasing importance of specifying the assumptions and goals of research. We argue that, to facilitate the organization and integration of study findings using diverse research strategies, it is critical to (1) specify the clinical construct(s) being investigated, (2) be transparent about and open to the study of divergent constructs, (3) address the construct validity of the specific assessment method(s) being studied, and (4) reconcile divergent approaches with the goal of deepening overall understanding of the psychopathy syndrome.

The Psychopathy Construct: A Historical Perspective

The notion of the psychopathic personality has been around for centuries (Hervé, 2017). Indeed, biblical, classical, and medieval texts describe individuals of societal consequence due to their poor behavioral control, stimulation-seeking, egotism, callousness, and predilection for rule violation that appears relatively refractory to punishment (e.g., Cleckley, 1941; McCord & McCord, 1964). While psychopathy itself has been established as a clinical disorder since the early 1900s, there have been debates and inconsistencies in its operationalization since that time (see Hare & Neumann, 2010). Indeed, in 1937 Braude claimed that "there is scarcely a theme in psychiatry more controversial and more in need of terra firma" (p. 238), noting the myriad of terms used to designate the classification of 'psychopath' ("moral insanity, constitutional immorality, constitutional psychopathic inferiority, constitutional psychopathy, sociopath, and psychosatipath;" p. 238). Similarly, Wilson and Persor (1939) noted that "the term psychopathic personality is applied to the members of an extremely heterogeneous group of 'half-crazy' individuals…Borderline psychotics, sexual perverts, hoboes, habitual drunkards, drug addicts, cranks, malingerers, criminals, misanthropes, and a host of other misfits… find themselves labeled constitutional psychopathic inferiors, or, less formidably, psychopathic personalities" (p. 122). Associated issues involving heterogeneity of the syndrome are also well-established. For example, Henderson (1947) argued for at least three psychopathy subtypes (i.e., "the aggressive", "the inadequate", and "the creative"), all of which were characterized by impulsive behavior, "primitive reactions" to social frustrations, and "deep feelings of impotence" in social situations. Alternatively, Karpman (1941) proposed that there were two types of psychopathy: symptomatic/secondary psychopathy, which he posited developed in response to environmental factors and psychological trauma and was characterized by emotional disturbance and neurotic conflicts, and idiopathic/primary psychopathy, which he felt was constitutional in nature and was characterized by a fundamental absence of emotional responses. Overall, while the social deviance and cost of psychopathy has long been recognized, until the mid-twentieth century there was little consensus as to what the term 'psychopath' denoted. As such, there was no unifying framework for what constructs were being measured and how they related to one another in terms of etiological mechanisms.

Cleckley's (1941) *The Mask of Sanity* was published against this background. In it, Cleckley set forth a set of sixteen criteria for what one reviewer would call "a *particular variety* of psychopathic personality" (p. 374, emphasis added, Lowrey, 1943), highlighting the ongoing disagreement regarding definition and heterogeneity. In *The Mask of Sanity*, Cleckley also addressed differential diagnosis, including distinguishing the psychopath from "the psychotic", "the psychoneurotic"[2], and "the ordinary criminal". He explicitly addressed the heterogeneity seen in the psychopathy syndrome and, critically, he proposed that there is a subset of psychopathic individuals, "traditionally confused with a fairly heterogeneous group under a loose and variously understood term" (p. 16, Cleckley, 1988) whose behavioral dysregulation could be explained by a discrete "defect or elimination which prevents important components of normal experience from being integrated into the whole human reaction" (p. 374). This subtype, referred to in this manuscript as the 'Cleckley psychopath,' is purported to be characterized by an inherent abnormality that underlies a diverse set of symptoms that can vary across social class,

[2] The psychopath described by Cleckley (1941) was one who appeared "incapable" of anxiety. Other authors (e.g., Karpman, 1948) differentiated this psychopath from those individuals with an underlying neurotic or psychotic personality, on top of which psychopathy symptoms had been layered. Since the PCL/-R does not effectively distinguish the neurotic/high-anxious psychopath from the low-anxious psychopath (Arnett et al., 1993; Lake et al., 2011; Newman et al., 2005; Schmitt & Newman, 1999), it is sometimes supplemented by measures of anxiety or neuroticism.

gender, and other individual differences variables. Cleckley specified that "the psychopath is not unnaturally pushed toward some sort of divertissement. Even weak impulses, petty and fleeting gratifications, are sufficient to produce in him injudicious, distasteful, and even outlandish misbehavior" (p. 389). In other words, while psychopathy may be expressed as excessive reward seeking, violence, or deviant sexual behavior, Cleckley clarifies that those behaviors are just manifestations of a core underlying deficit.

While reviews of the original book and its revised edition praised Cleckley's thorough review of psychopathy theory and the extensive case descriptions (Healy, 1942; Sumner, 1941), there was a more ambivalent response to Cleckley's account of psychopathic dysfunction. Cleckley's proposal was described as a "somewhat inconclusive theory" (p. 904, Sumner, 1941) by an author who "shows a somewhat cavalier attitude toward the proposals of other authorities" (p. 88, Macdonald, 1966). However, other reviewers argued that this proposed deficit nevertheless provided "one of the best examples available of careful clinical observation leading to hypotheses (not conclusions) which can be put to controlled test" (p. 197, Maher, 1973) and spurred interest in empirical investigations of the deficit and the psychobiological factors that might underlie it. The goal of the construct validation process is to specify a construct and evaluate models to expand understanding of that construct through exploration of etiological correlates and other components of the nomological network. Cleckley's emphasis on explaining psychopathy as resulting from a fundamental deficit in cognitive and emotional processing was an important aspect of his work that has driven subsequent attempts to isolate the psychobiological mechanisms that underlie the condition (e.g., Hare, 1978, 1979; Patterson & Newman, 1993).

In the years following Cleckley's first publication, researchers continued to rely primarily on heterogenous criteria and labels for classifying psychopathic individuals, which resulted in conceptual and empirical difficulties in the field. Specifically, researchers utilized either a generic psychiatric diagnosis that sometimes included checklists derived from existing clinical descriptions (e.g., Bryan & Kapche, 1967; Gilbert, 1945; Rimé et al., 1978; Steinberg & Schwartz, 1976; Thorne, 1947) or a more measure-driven assessment approach, with the syndrome being operationalized by scores on assessment instruments including the 16PF (Gorman, 1970), or, most commonly, the pd subscale of the MMPI (Benton & Probst, 1946; Doctor & Craine, 1971; Heilbrun, 1979; Hetherington & Klinger, 1964; House & Milligan, 1976; Rubin, 1948; Whitelock et al., 1971). In an examination of the intercorrelations between various measures of psychopathy, including the 16PF and the MMPI,, Hundleby and Ross (1977) did not find sufficient agreement amongst the scales to suggest that they assessed a uniform construct, and concluded with a warning that if researchers selected their psychopathic and non-psychopathic on the basis of one of these instruments, "the likelihood is low that a different measure would yield the same arrangement of subjects" (p. 703). As such, it would still be argued as late as the mid-1970's that "psychopathy is an undefinable and untreatable general form of antisocial personality disorder" (p. 283, Rotenberg, 1975).

Introduction of The Psychopathy Checklist

During this period, a relatively small group of researchers were relying heavily on a classification system based on the Cleckley criteria to test hypotheses involving cognitive and psychobiological processes associated with psychopathy (Hare, 1972, 1979; Hare et al., 1978; Raskin & Hare, 1978; Schalling & Rosén, 1968; Widom, 1976), a more consistent approach that better facilitated comparisons across studies. To address both the continued heterogeneity of the field and the challenges of this methodology, Hare (1980) developed a 22-item scale to assess psychopathy based on the Cleckley criteria. Relative to other self-report and behaviorally based assessment measures, this scale was unique in two key ways: (1) in tapping the Cleckley psychopath, it provided a tool for tapping into the purported Cleckley deficit, and (2) it focused on capturing the diverse manifestations of the Cleckleyan defect. The initial checklist would be revised and eventually published as the PCL/-R (1991a). The PCL/-R "brought a semblance of order to the field" (p. 449, Hare & Neumann, 2010) by allowing reliable measurement based on a construct that until that point was elusive.

In sharp contrast to the heterogeneity in psychopathy assessments of the mid-century, in the decades following the publication of Hare's checklist, strong consensus emerged around the instrument. The number of articles referencing the PCL/-R rose from approximately 20 between 1991 and 1995, to over 150 between 2001 and 2005, and nearly 250 between 2006 and 2010. This steadily increasing interest in the PCL/-R over this period had major impacts on the field (Hare, 1996; Hare & Neumann, 2008), particularly in terms of its utility as a predictor of criminal recidivism and violence (DeMatteo & Edens, 2006; Dolan & Doyle, 2000; Hemphill et al., 1998). Two other impacts are not as often explicitly noted: first, the growing use of the PCL/-R indirectly reified Cleckley's conceptualization of psychopathy. To the extent that the PCL/-R was developed to reflect Cleckley's clinical conceptualization (Hare 1980, 1991a), whenever the instrument was used as the operationalization of the construct, it was Cleckley's construct that was being assessed. Second, in part because of its connection to Cleckley's construct, the PCL/-R fostered a deficit-based conceptualization of

psychopathy and facilitated the development of mechanistic explanatory models of the syndrome. The PCL/-R framed psychopathy as a clinical construct, which encouraged etiological models based on abnormalities in psychobiological, cognitive, and emotion processes (Hare, 1996). Importantly, Hare demonstrated that the PCL/-R predicts the Cleckley criteria and demonstrated the validity of the assessment measure for Cleckley's construct.[3] Moreover, the rating scale has been shown to be reliable when used by both established and beginner raters (Cooke and Michie 1997; Hare 1991b, 1998, 2003; Hare et al. 1990, 2000). These data cement the PCL/-R's utility as the operationalization of the theoretical Cleckley psychopath.

Influence of the PCL/-R on Mechanistic Theory Advancement: The Response Modulation Hypothesis (RMH)

Part of the process of construct validation involves specification of intervening mechanisms that are relevant to the construct of interest (Whitely, 1983; see also Craver & Tabery, 2023; Grahek et al., 2021 for further discussion of explanatory mechanistic models in psychological science). One mechanistic model of psychopathic dysfunction that benefited greatly from the conceptual and assessment clarity of the PCL/-R was the response modulation hypothesis (RMH); this theoretical model strove to elucidate mechanistic underpinnings of the regulatory deficit inherent to the Cleckley psychopath, thereby expanding the nomological network around the construct. The RMH has its roots in early work by Gorenstein and Newman (1980), who were developing models to explain a diverse collection of syndromes of disinhibition, one of which they associated specifically with the Cleckley psychopath, but all of which they argued were associated with a shared, underlying psychobiological mechanism analogous to dysfunction in the medial septum, posterior hippocampus, and orbito-frontal cortex (Newman et al., 1983).

Between 1980 and 1993, Newman and colleagues developed a model of psychopathy based on the inability of the psychopathic individual to engage efficient response modulation (i.e., the ability to alter a dominant response set in response to secondary stimuli). Because of the established association between psychopathy and passive avoidance deficits (e.g., Blackburn, 1983; Lykken, 1957; Schachter & Latané, 1964), early tests of this model utilized go/no-go paradigms involving monetary incentives. For example, Newman and Kosson (1986) showed that the passive avoidance deficits associated with psychopathy were contingent on task demands. In their study, psychopathic participants performed as well as controls in a condition that involved only the avoidance of punishment but performed significantly worse than controls in a condition that involved competing reward and punishment contingencies. These and other results challenged the idea that psychopath's poor performance on passive avoidance tasks was the result of a general insensitivity towards punishment or oversensitivity to reward, and instead tied these performance deficits to competing task demands. However, Newman and colleagues (1990) also showed that, when the go-no-go paradigm was modified such that both reward and punishment contingencies were salient from the start, performance differences between groups were erased. Similarly, using an alternative paradigm that involved a steadily increasing probability of punishment following the establishment of a dominant response for reward, Newman et al. (1987) found that psychopathic individuals initially failed to modify their response in the light of the changing contingencies. However, when response feedback was followed by a five-second waiting period, the response perseveration associated with psychopathy was eliminated.

Based on these and other findings (Howland et al., 1993; Newman et al., 1985, 1987, 1990) and building on the Gorenstein and Newman (1980) septal model, Patterson and Newman (1993) proposed a psychological mechanism for syndromes of disinhibition rooted in the four stages of response modulation (i.e., the establishment of a dominant response set, the reaction to an aversive event, the resulting behavioral adaptation, and the effects of reflection). The authors proposed that deficits in the response modulation process represent a shared common factor across syndromes of disinhibitions (e.g., hyperactivity, alcoholism, psychopathy, extraversion). However, they also proposed syndrome-specific deficits. For example, whereas the response modulation deficits observed in some cases might arise from over-sensitivity to reward cues or under-sensitivity to punishment cues, in psychopathy, the deficit was argued to be associated with a failure to suspend an established dominant response and deploy attentional resources to processing the consequences of their responding.

The RMH continued to develop throughout the 1990's and 2000's. One of the most impactful findings during this period involved the use of a picture-word paradigm to test the hypothesis that psychopathic individuals' response modulation deficit was not specific to fear or punishment-related stimuli (Newman et al., 1997). In this task, participants were asked to determine if two pictures presented consecutively

[3] Ratings of psychopathy based on Cleckley's 16 original criteria and PCL-/R total scores are strongly correlated, with the association between Cleckley's criteria and the total score stronger than the relationships with either PCL/-R Factor score independently. This association is true for both male (Hare, 1980) and female (Kennealy et al., 2007) samples (both have $r = 0.83$).

were related (e.g., a picture of a fork and a picture of a spoon) or if two words presented consecutively were related (e.g., the word "smoke" and the word "cigarette"). Importantly, the initial stimulus, termed the 'context display,' included both a line drawing and a superimposed word. For each trial, participants were instructed to focus on either the word or the picture to determine its relatedness to the stimulus on the upcoming 'test display.' For example, on a picture (P) trial, participants were to focus on the line drawing on the context display to determine its relatedness to the line drawing on the subsequent test display while ignoring the superimposed word on the context display. Critically, the stimuli on the test display sometimes related to the to-be-ignored word or line drawing on the context display. For example, on a P trial, a participant may be shown a picture of a sandwich, with the superimposed word "sweep" on the context display and then shown a picture of a broom on the test display. Because the participant is supposed to focus on the picture only, the correct answer in this case is "unrelated". However, research showed that participants experience interference on these trials, relative to comparison trials where the to-be-ignored stimulus is unrelated to the test display (Gernsbacher & Faust, 1991). Based on the RMH, Newman et al. (1997) speculated that, once they were engaged in goal directed behavior (i.e., focusing on the primary stimulus on the context display), psychopathic participants would show less interference than control participants on the experimental trials. As predicted, psychopathic participants with low anxiety scores (i.e., "primary psychopaths") did not exhibit the normally robust interference effect. This result was consistent with the predictions of the response modulation hypothesis, which held that psychopathic individuals would be less affected by secondary cues when they are engaged in goal-directed behavior.

This finding was particularly important for two key reasons: first, it showed that psychopathic individual's deficits in performance could be observed on tasks where the stimuli were affectively "neutral" (i.e., the stimuli were not explicitly reward- or punishment-related), an important finding given the field's longstanding emphasis on emotion stimuli and emotion-based explanations for psychopathy. Second, on the picture-word task, it was predicted that abnormalities in psychopathic individual's performance would be represented by relatively better performance on the task (i.e., showing less interference). As a result, the psychopathic individuals' performance on this task was difficult to attribute to an absence of performance motivation.

Other tests of the response modulation hypothesis continued to focus on elucidating the nature of the proposed deficit in the psychopath's proclivity to suspend a dominant response set in the presence of secondary stimuli and then use that information to modulate behavior. These studies utilized a variety of paradigms that involved the processing of primary and secondary cues, including emotional lexical decisions tasks (Lorenz and Newman 2002a), response competition flanker-type tasks (Zeier et al., 2009), conflict monitoring tasks (Zeier & Newman, 2013), and a separated color-word Stroop task (Hiatt et al., 2004). Given the affectively neutral aspects of many of these tasks, it was difficult to account for the findings using only an emotion (e.g., fear)-based deficit model of psychopathy. Consistent with this deficit and in keeping with the hypothesized attentional mechanisms of the RMH, Larson et al. (2013) showed that abnormalities in amygdala activation exhibited by psychopaths and typically associated with low-fear or emotion-based explanations for psychopathy (Blair, 2009; Lykken, 1957) could also be moderated by goal-directed attention. When experimental conditions required psychopathic participants to attend to threat cues, amygdala activation did not differ between psychopaths and nonpsychopaths; rather, it was only under conditions where their attention was engaged by a goal-relevant task that psychopaths showed diminished amygdala activation in response to subsequent threat cues (see also replications of this basic phenomenon that examined fear potentiated startle; e.g., Baskin-Sommers et al. 2011a, b, 2013). Taken together, these studies highlight that classic psychopathic deficits in affective processing may be eliminated via attentional manipulation, which highlights the importance of attentional factors when considering the etiology of psychopathic dysfunction. Critically, the ability to make a key causal deficit disappear based on theoretical understanding has direct implications for the treatment and management of the disorder.

From almost the earliest days, tests of the RMH were rooted in the Cleckley conceptualization of psychopathy. Like Cleckley, the proponents of the RMH argue that there is at least one clinically meaningful subtype of what we call "psychopathy" that is characterized by a fundamental deficit in information processing that does not reflect extreme variations on normal personality and that can be used to explain these individual's maladaptive interpersonal and emotional functioning (Wallace et al., 1999). Because of its close connection to the Cleckley criteria, the PCL/-R became the instrument of choice for testing the model. In contrast to the problems in the years before the emergence of the PCL/-R, it could now be assumed that the high and low scoring participants were the same across these studies. While not every test of the RMH provided support (e.g., Dvorak-Bertsch et al., 2007; Lorenz & Newman, 2002b), the continued use of the PCL/-R and its emphasis on a unified psychopathy construct enabled the positive and negative results to be integrated and for the model to be continuously revised.

In considering the process of construct validation as it relates to exploration of the etiological underpinnings of the

theoretical Cleckley psychopath, it is important to recognize that Newman and colleagues modified sample selection criteria to allow more valid assessment of the construct of interest. Newman (1998) noted that the PCL/-R does not capture Cleckley's exclusion criteria. As such, he "argued that researchers wishing to study hypotheses based on Cleckley's work should make use of additional measures to refine diagnoses made using the PCL/-R" (p. 84, Brinkley et al., 2004). Specifically, he advocated for inclusion of measures of anxiety, intelligence, and psychosis to identify individuals with high anxiety, low intelligence, and psychosis and exclude them based on Cleckley's criteria that psychopathy is characterized by "absences of nervousness," "good intelligence," and "absence of delusions." This modification of selection criteria based on theoretical rationale (i.e., removing individuals from the sample with a presumed alternative mechanism underlying shared observable behavior) serves as a model of construct validation in that the adjustment allows for more focused assessment of the variable of interest (the purported Cleckley psychopath deficit). Newman's incorporation of exclusion criteria allowed for more refined identification and isolation of the Cleckley psychopath, which in turn allowed for more refined testing of a shared mechanism of dysfunction.

Beyond allowing for the formulation and systematic testing of the RMH, the PCL/-R has facilitated the formulation of derivatives and/or alternative mechanistic accounts born of or inspired by the RMH. Moreover, it enabled the ability to contrast predictions by providing a standard and reliable assessment of psychopathy. One derivative approach has posited that the deficits associated with psychopathy are the result of early selective attention processes. According to some theorists, the early stages of selective attention constitute a "fixed bottleneck" that can lead to a blocking of goal-irrelevant information (Driver, 2001). In this context, the failure of psychopathic individuals to reallocate attention to potentially meaningful cues once a dominant response is engaged is the result of dysfunction at this early, bottleneck stage of selective attention (Baskin-Sommers, Curtin et al., 2011; Baskin-Sommers et al., 2013; Hamilton et al., 2014; Krusemark et al., 2016; Wolf et al., 2012). A second derivative of the RMH contextualizes psychopathic deficits within a framework of cerebral asymmetries and interhemispheric transfer. Given findings that suggest the deficits exhibited by psychopathic individuals are especially apparent when there are high demands on left hemisphere processing (e.g., Bernstein et al., 2000; Hiatt et al., 2004), Kosson (1996) proposed the left-hemisphere activation (LHA) hypothesis, which states that psychopaths' response modulation deficits are specific to conditions that place substantial demands on left-hemisphere-specific processing resources (Kosson, 1998). For instance, psychopathic individuals show reduced accuracy on a divided attention task involving the presentation of lateralized stimuli when the majority of stimuli are presented in the right visual field (thereby disproportionately taxing left hemisphere resources) versus when they are presented equally across both visual fields (Llanes & Kosson, 2006; see also Kosson et al., 2007). The LHA therefore attempts to further specify conditions under which psychopathic information processing abnormalities occur.

In addition to research focused on more fully explicating the RMH deficit, the RMH also inspired the Impaired Integration (II) theory, which takes the cognitive, affective, and behavioral deficits in psychopathy, many of which are associated with the RMH, and places them in a brain-based framework (Hamilton et al., 2015). It proposes that psychopathy is characterized by an integrative deficit that undermines the ability to rapidly accommodate multi-channel information. Specifically, the II theory postulates that psychopathic individuals have a perceptual bottleneck that undermines their ability to rapidly integrate multicomponent sensory information that results in a sequential information processing style, which in turn impacts associative processing as well as neural network topography and coordination. Consistent with this theory, psychopaths show deficits in simultaneous but not sequential processing (Hamilton & Newman, 2018). Further, the II theory was the first theoretical framework of psychopathy to situate psychopathy in a systems-based neural context. It therefore accommodates both behavioral and neurobiological findings (e.g., abnormal or reduced neural connectivity or coordination in functional and/or structural networks) in this population (Anderson et al., 2017; Contreras-Rodríguez et al., 2015; Deming et al., 2023; Espinoza et al., 2019; Lindner et al., 2018; Philippi et al., 2015; Sheng et al., 2010; Tillem et al., 2018; Yang et al., 2012).

Taken together, the PCL/-R offered a framework for identifying a subset of high psychopathy individuals with a shared underlying deficit, allowing the elucidation of an etiological network for understanding psychopathy. Notably, other investigators have also employed the PCL-R to explore etiological correlates of Cleckley psychopaths, with good results (e.g., Blair et al., 2002; Craig et al., 2009; Decety et al., 2014).

Prevailing Criticisms of the PCL/-R

While the PCL/-R has been influential in clinical and forensic settings and has provided a standardized, highly reliable measure to identify a group of Cleckleyan psychopaths that exhibit a shared underlying deficit, it has not been without critique. In recent years, there has been growing criticism of the utility of the PCL/-R as a diagnostic tool as well as a gradual shift from using this measure as it was originally

designed, that is, to designate the prototypical Cleckleyan psychopath. While critical evaluation of the measure is an important and necessary part of the construct validation process, failure to conceptualize results within the context of the measure's original design and/or disregard of the measure altogether can threaten this process. There are three primary critiques of the PCL/-R. The first criticism is that it yields a heterogenous group of individuals all labeled 'psychopaths' whose psychopathic traits reflect different etiologies (see Skeem & Cooke, 2010). The second critique focuses on the notion that reliance on the PCL/-R total score to operationalize of psychopathy risks overlooking core constituent dimensions of psychopathy (e.g., its factor structure or assessment of the construct through different measures). This critique purports that to understand diverse forms and aspects of psychopathy, researchers should "focus on the etiology of the diverse personality traits constituting psychopathy and learn how these traits combine or interact to give rise to the full clinical picture of psychopathy" (p. 605, Lilienfeld et al., 2015; see also Patrick, 2022). The third criticism is that the PCL/-R has been used as the singular operationalization of psychopathy and that, as a result of this 'mono-operation bias', research findings may be specific to the measure itself rather than the syndrome.

Critiques of the PCL/-R Reconsidered

While these critiques have important implications for psychopathy research, specifically the need to (a) recognize etiologically distinct subtypes, (b) incorporate alternative ways of conceptualizing psychopathy, and (c) employ alternative assessments to evaluate these various constructs, the addition of these alternative approaches should not supplant successful research programs that use the PCL/-R to study the prototypical Cleckley psychopath. Given its success in reliably identifying a prototypic psychopathy syndrome, predicting the essential clinical symptoms of the disorder, and characterizing distinctive and etiologically relevant deficits capable of informing early identification and treatment of the disorder, there are compelling reasons for continued use of the PCL/-R until a better measure of this syndrome is developed.

Notably, the criticisms outlined above do not actually call into question the validity or utility of the PCL/-R as a measure of Cleckley psychopathy. Considering the first critique, the heterogeneity of the psychopathy syndrome has been a topic of discussion since the mid-twentieth century, and since that time there has been good progress delineating psychopathy subtypes (Blackburn, 2009; Hervé, 2007; Hicks & Drislane, 2018; Lykken, 2006; Porter, 1996). In fact, Cleckley (1941) made clear that the psychopath whose behavioral dysregulation could be explained by a discrete "defect or elimination which prevents important components of normal experience from being integrated into the whole human reaction" (p. 374, 1988) was only a subset of what are called psychopathic individuals. As outlined in earlier sections, the PCL/-R critically provided the framework to allow the Newman lab to systematically isolate and study a subset of high psychopathy individuals who were purported to share a common underlying deficit (see Brinkley et al., 2004). To this end, Newman and colleagues focused on participants with uniformly high PCL/-R scores[4] (i.e., dysregulation in multiple domains) and, based on Cleckley's proposed exclusion criteria, omitted individuals with high levels of anxiety/negative affect, psychoses, or low intelligence. Thus, appeals to move away from the PCL/-R ignore its utility in isolating the subtype of psychopath outlined by Cleckley and in studying the relatively rare, abnormal predisposition (i.e., core deficit) underlying their 'psychopathic behavior.'

According to criticisms, use of PCL/-R total scores is also problematic because it fails to address more specific traits and dimensions associated with psychopathic behavior (e.g., self-centered impulsivity, cold-heartedness, fearless dominance, boldness, meanness). This second critique makes a valid point but appears to ignore the original goal of the PCL/-R: to identify *Cleckley* psychopaths and outline behavioral criteria that serve as imperfect indicators of a shared mechanism of dysregulation. The fact that it does not readily facilitate the study of more specific traits does not diminish its ability to achieve its original aims.

Regarding the third critique that findings may be specific to the measure when focusing on the PCL/-R alone, we believe that exploring the correlates of psychopathy identified using alternative, trait-focused methods will advance psychopathy generally, but may not be as well suited to capturing Cleckleyan psychopathy. Cleckley (1941) chose not to define his psychopathic subgroup based on a uniform set of traits (e.g., fearlessness, antagonism, criminality). Rather, he developed criteria composed of indicators that captured an underlying inability to experience the world as others do (e.g., pathological egocentricity and incapacity for love) and to regulate responses (e.g., poor judgement and failure to learn from experience), which he argued stemmed from a shared profound defect. Indeed, in his case studies, Cleckley captures the heterogeneity of traits seen in high psychopathy individuals. In one example, Cleckley describes 'Max,' a boisterous and boastful male veteran of small stature who in initial interactions was congenial and accommodating but quickly demonstrated restlessness, arrogance, and verbal aggression. He asserted far-fetched personal information

[4] In the Newman lab, the proposed PCL/-R total score value of 30 or above has been highly effective capturing the Cleckley psychopathy.

about his medical history and/or achievements and had an established pattern of instigating altercations with others; while psychiatrically hospitalized he even orchestrated fights between other residents and sowed seeds of unrest and rule breaking among the patients. In a second case, Cleckley discusses 'Pierre/Pete,' a young, intelligent college student from a respected Southern family who forged multiple checks purely on impulse despite the potentially serious ramifications of the action for his academic career and legal status and freely spoke of engaging in other delinquencies that he carried out for no reason other than a whim. In another case example he describes 'Anna,' who appeared on the surface as an engaging, charming, and intelligent woman from a wealthy upbringing. Upon conversation she freely discussed her myriad non-discriminant and haphazard sexual encounters throughout her life and demonstrated a thoughtless belief that she was trustworthy and honorable despite extensive evidence of repeated lying, delinquent behavior, and damaged interpersonal relationships requiring her family's support for treatment and assistance with legal run-ins. In his accounts, Cleckley highlights how each of these individuals appear to have an innate "defect" such that, "despite a high capacity for intelligence and charm, something necessary for wisdom or for sincere and major human feelings was left out" (p. 116, 1988), despite differences in presentations.

Through our research assessing thousands of prisoners using the PCL/-R, we have been struck by the dramatically different trait profiles displayed by individuals with equally elevated PCL/-R scores (i.e., >= 30). Indeed, solely based on gender alone there are stark differences in traits observed in high psychopathy individuals (e.g., de Vogel & Lancel, 2016). From a mechanistic standpoint, we can understand psychopathic symptoms (e.g., interpersonal, affective, behavioral, lifestyle) in high psychopathy individuals as manifestations of a core underlying process, with symptom expression shaped by external (e.g., situations, income, childhood experiences) and internal/biological (e.g., gender) contexts. An additional benefit of the mechanism-focused approach is that it allows consideration of how personality/temperament interacts with a fundamental deficit. Given that the expression of psychopathy varies across individuals, as exemplified in Cleckley's examples, we are persuaded to follow Cleckley's example and look beyond specific trait clusters towards a more abstract shared defect of interest.

Skeem and Cooke (2010) warned against the temptation to reify PCL/-R defined psychopathy. That is, the measure should not become synonymous with the construct. Though we agree with this concern, the PCL/-R was developed and validated as a measure of Cleckley psychopathy. For this reason, it is particularly well suited to address Cleckley psychopathy in its traditional form. Though more recent approaches to psychopathy have attempted to improve upon the construct and use divergent methods to assess the construct in alternative and perhaps more sophisticated ways, we are concerned that these approaches have muddied the water rather than clarifying the construct validity associated with the Cleckley psychopath. To clarify this claim, we will discuss two popular and well-developed alternatives to the PCL/-R-based approach advocated in this paper: Lykken's low-fear hypothesis (1957, 1995) and the Five Factor model of psychopathy (FFM; Widiger & Lynam, 1998; see also Brinkley et al., 2004; Lynam & Widiger, 2007).

Relation of Lykken's Low-Fear Model to Cleckley Psychopathy

Although Lykken initially embraced Cleckley's writings on psychopathy, he argued that Cleckley's theoretical formation was too vague to guide experimental research and that "a simpler hypothesis was both adequate and testable" (p. 134, Lykken, 1995). He proposed that researchers substitute the low fear construct for Cleckley's characterization of the core defect in psychopathy (see the 'Improving the Diagnosis of Psychopathy' in Lykken, 1957). His proposal facilitated research on psychopathy and generated support for his proposal that psychopathic individuals are characterized by low fear (see reviews regarding this support, e.g., Patrick, 2017). While reviewing progress on the low-fear model in his book *The Antisocial Personalities* (1995), Lykken further proposed that research on primary psychopathy could be improved by focusing on Factor 1 of the PCL/-R (PCL-1, see Hare et al., 1990) and/or supplementing the PCL/-R with measures that target fearfulness and a fear-related deficit more directly, namely "harmavoidance." He wrote "if the low-fear hypothesis is valid, then the Cleckley or PCL-1 nominated psychopaths who are low in harmavoidance should constitute the purist group of primary psychopaths, whereas the prison inmates who are not psychopathic according to the PCL-1 and who are high on harmavoidance should provide the best contrast" (p. 127). In contrast to his earlier proposal, which guided the selection of dependent variables in psychopathy research (e.g., skin-conductance, fear conditioning, affect-modulated startle, amygdala activation), Lykken's latter proposal facilitated a shift in the selection of independent variables, such that tests of the low-fear hypothesis began focusing on Factor 1 of the PCL/-R or other fear-related dimensions. As expected, support for the low-fear hypothesis proved more consistent when researchers focused on Factor 1 as opposed to Factor 2 (i.e., impulsive antisocial component) of the PCL/-R. Indeed, results for Factor 2 were typically nonsignificant and sometimes in

the opposite direction from Factor 1 findings, with results for PCL/-R total scores falling somewhere in between those for the two factors. Similar results were found when investigators employed self-report measures of fearlessness (e.g., harmavoidance or fearless dominance) rather than Factor 1 of the PCL/-R.

When evaluating the above findings, it is crucial to consider whether, in changing the assessment criteria, researchers shifted from evaluating the Cleckley construct to an alternative psychopathy subtype. Put another way, the modification of assessment criteria may have resulted in the capture of a hypothetical construct distinct from the Cleckley psychopath. On one hand, Lykken noted that he was modifying Cleckley's proposal on the etiology of psychopathy. On the other hand, he continued to address the Cleckley psychopath, noting that his model could explain all the symptoms described by Cleckley and that assessment of the Cleckleyan (or 'primary') psychopath could be better achieved using PCL/-R Factor 1 and self-report measures of fearlessness (see Lykken, 1995). Given this lack of clarity, one interpretation of the stronger findings obtained when focusing on PCL/-R Factor 1 is that measures of psychopathy like the PCL/-R may be improved by placing greater emphasis on fear-related traits/components. This interpretation presumes that stronger associations with low-fear outcome measures indicate superior construct validity, even though low fear is not a salient aspect of Cleckley's psychopathy construct. When participants identified by high Factor 1 scores or self-report measures of fearlessness display low fear on some dependent measure, the results support the construct validity of the fearlessness measures and the use of Factor 1 as a fear proxy. However, they do not necessarily support the low-fear model of *psychopathy* or demonstrate that PCL-1 and harm avoidance are better measures of Cleckley psychopathy than the PCL/-R.

As already discussed, Cleckley postulated the existence of a unitary cognitive/affective deficit as the key factor explaining the diverse behavior problems associated with psychopathy. This prediction clearly extends to Factor 2 as well as Factor 1 items of the PCL/-R. Thus, any failure of Factor 2 items to perform like Factor 1 items in predicting low-fear deficits would seem to contradict the idea that low fear represents a superior conceptualization of the Cleckley deficit or that PCL-1/fearless measures are superior to PCL/-R total scores for measuring Cleckley psychopathy. The point here is that it is crucial to specify what theoretical construct is being investigated and relate findings to the relevant constructs. Regarding the focus of this manuscript, we believe that findings obtained for specific psychopathy components but not for the *entire construct* have minimal implications for the construct validity of the PCL/-R or for proposals relating response modulation models to the Cleckley construct.

Aside from efforts to maximize correlations between low fear and psychopathy, the transition to analyzing PCL/-R factors and facets has been facilitated by factor analytic studies that highlight the multifactorial nature of the PCL/-R. These findings have been interpreted to imply that psychopathy might be assessed and understood more accurately using measures of the partially independent factors or facets of psychopathy. Moreover, some investigators have interpreted the multi-factorial nature of the PCL/-R as a need to specify multiple etiological processes. However, owing to the substantial overlap among the factors and facets, others have argued that psychopathy may be accurately understood as a unitary construct. This view purports that differences in the factor and facet scores reflect diverse manifestations of a specific underlying deficit rather than diverse causes of the psychopathy syndrome (see Hamilton et al., 2015). The existence of these divergent views begs the question of whether psychopathic traits reflect different causes that are being combined, whether they reflect a single cause being multiply expressed or a combination of these processes. It is worth noting that past research has cautioned against how the act of partialing variables might impact the interpretation of theoretical constructs. Specifically, the practice of partialing "runs the danger of losing sight of the original construct under investigation or of changing the nomological network" (p. 234, Hare & Neumann, 2008). Put another way, partialing variables can distort the construct such that the remaining factor is distinct from the original latent construct (Hoyle et al., 2023; Lynam et al., 2006). In terms of psychopathy specifically, this relationship is clear in that the PCL/-R factors have unique correlates relative to the PCL/-R total score (Choy et al., 2022; Edwards et al., 2023; Hauser et al., 2021; Hofmann et al., 2024; Korponay & Koenigs, 2021; Sandvik et al., 2014; Smith & Newman, 1990; Sohn et al., 2022; Yoder et al., 2022). Studying PCL/-R factors at the expense of a total score, which was designed to capture the entirety of the Cleckley psychopath, increases risk of conceptual errors. Sole focus on the isolation of interactive traits that comprise psychopathy undermines identification of causal mechanisms shared among the traits that could be targeted for intervention and risks distortion of the construct validation process. The same argument can be made for the study of callous-unemotional traits and triarchic dimensions: while study of these constructs is important, these constructs should not be conflated with the Cleckley psychopath without evidence of convergent and divergent validity.

Some research efforts have focused on the study of relative contributions of high F1 versus high F2 scores on mechanistic correlates. While this approach might offer

interesting insights into mechanisms underlying each of these individual traits, we argue that high F1 or F2 *in the absence of high scores on the other factor* is not Cleckleyan psychopathy. Given that there is far less variance between the factors when PCL-R total scores are >=30 relative to mid-range scores, we suspect that high PCL-R scores driven predominantly by one factor over the other reflect a different mechanism than that underlying the true Cleckley psychopath. Indeed, throughout Cleckley's work he argues that the traits comprising F1 and F2 are inextricably linked, driven by the same inherent deficit. While the Newman lab has reported some results contrasting high F2 alone versus high F1 and high F2 groups in several instances, this was not intended to examine the specific meaning of the factors but rather to compare a psychopathic group (high overall scores on the PCL) with an externalizer group (impulsive / antisocial traits without concomitantly high scores on F1). For the field to advance, clarity in the clinical construct(s) of interest, transparency and openness to the study of divergent constructs, and construct validation efforts are critical to facilitate the organization and synthesis of study findings using diverse research strategies.

Relation of the Five-Factor Model to Cleckley Psychopathy

In contrast to the Cleckley and Lykken models of psychopathy, the Five-Factor model (FFM; Widiger & Lynam, 1998) seeks to characterize psychopathy using well-established, normal personality dimensions. More specifically, the model relates psychopathy to low Agreeableness, low Conscientiousness, high Extraversion, and a combination of low and high Neuroticism (Derefinko & Lynam, 2013; Lynam & Miller, 2015), with further differentiation available using the facets of these dimensions. Owing to the range of Factors and facets, which were derived to characterize all aspects of human behavior, the FFM is well-suited to addressing the heterogeneity of psychopathy, its subtypes, and its constituent traits. For instance, investigators may distinguish primary (i.e., Cleckley) and secondary psychopathic subtypes using low and high levels of Neuroticism. Factor 1 and 2 of the PCL/-R may be distinguished by their relative levels of antagonism and conscientiousness. Thus, the FFM may be used to characterize existing psychopathy types or identify alternative types which, in turn, may be studied to clarify their respective construct validities. As with other approaches to psychopathy, FFM may be used to study psychobiological mechanisms by examining the correlates of FFM-specified psychopathic groups or dimensions. Given the near infinite variety of psychopathy-related types and traits that may be identified using the FFM, this approach is well suited to expanding the purview of the psychopathy field as recommended by Skeem and Cooke (2010).

According to the view that psychopathy must be understood as a combination of diverse traits, it is essential to specify these traits and examine their nomological networks (i.e., address the construct validity of the various types). Investigators seem to assume that components of psychopathy may be studied and that the findings obtained for any psychopathy-related trait will automatically be relevant for the overall psychopathy construct/diagnosis, as if the correlates of a component will always be the same regardless of psychopathy-level. However, this is a questionable assumption in need of specific investigation. Just as high psychopathy scores may reflect diverse subtypes, multiple dimensions, and distinct causal processes, there are numerous pathways to each specific trait (e.g. FFM dimensions, harm avoidance, Factor 1 scores, etc.). Such considerations raise the possibility that findings obtained for isolated traits or factors may not apply to the psychopathy construct at large, which itself involves a relatively specific context (i.e., high scores across *all* psychopathic traits). In other words, it is possible that the correlates of trait dimensions like openness, extraversion, or neuroticism found at low levels of these dimensions or in the context of relatively low PCL/-R total scores may not generalize to individuals manifesting high scores on all facets of psychopathy. As already discussed, it was the combination of problems seen across multiple domains (i.e., all facets of the PCL/-R) that led Cleckley to postulate a general cognitive / affective defect in psychopathy. Consistent with Pujara and colleagues'(2013) assertion that the neurobiological correlates of PCL-R scores of >=30 do not appear continuous (i.e., there are distinct neurobiological correlates seen in high scoring individuals that are not seen in lower scoring individuals; see Koenigs et al., 2011), it could be that elevations on psychopathy-related traits such as impulsivity in the context of low overall psychopathy levels reflect different causal processes than the same psychopathy-related traits (e.g., impulsivity) in the context of high overall psychopathy. Whereas high impulsivity is often associated with high reward sensitivity in the general population, Cleckley's perspective on psychopathy suggests that the impulsive behavior of psychopathic individuals reflects a different, more general, dysfunction that underlies the full range of psychopathic traits. Overall, careful interpretation of the relationship of trait-related findings to other empirical correlates is necessary to facilitate interpretation.

We wish to be clear that our concerns regarding the use of PCL/-R Factors and facets and FFM personality traits are specific to the study of *Cleckley* psychopathy. Trait-focused approaches to psychopathy have become as common, if not more common, than investigations of the categorical diagnosis and have yielded consistent and important results.

Thus, as advocated by Skeem and Cooke (2010), the use of alternative approaches to psychopathy has broadened our understanding of psychopathy-related traits.

Refining Theoretical Constructions through Attention to Mechanistic Correlates

We believe it is crucial to focus on etiologically relevant mechanisms in addition to symptoms, trait descriptors, and antisocial behaviors when validating diverse psychopathy-related constructs. By focusing on mechanisms, it is possible to address the problematic heterogeneity that exists in psychopathy and psychopathy-related dimensions. It is well-established that the understanding of causation and causal mechanisms critically informs models of clinical phenomena (Thagard, 1998). We advocate for inclusion of and focus on mechanistic correlations to facilitate the refinement and study of constructs of interest, whether it is Cleckleyan psychopathy or another variant. Investigators may be able to demonstrate that the response modulation deficits associated with the PCL/-R are also found when identifying psychopathic samples using the FFM. Similarly, investigators may discover that the low-fear correlates associated with PCL/-R Factor 1 and harm avoidance are also reliably associated with PCL/-R total scores. By focusing on mechanisms, it is also possible to demonstrate that novel subtypes of psychopathy have been identified and that other aspects of the construct validity (e.g., treatment response) align with the distinctive mechanisms. Furthermore, clarifying the etiological process underlying the various traits would facilitate exploration of trait interactions relevant to psychopathy.

Additionally, inclusion of mechanism-focused approaches to construct validation enables bottom-up manipulation of observed traits and behaviors (Bechtel, 2009; Craver, 2007; Craver & Bechtel, 2006; Glennan, 2005). Construct validation of clinical entities ultimately relies on identifying the disorder via its symptoms and specifying distinct mechanisms that are useful for early identification and treatment of behavior problems (Morey, 1991; Smith et al., 2009). For psychopathy, there is emerging literature that different antisocial subtypes respond differentially to various treatment approaches. Using a mechanism-focused approach, Baskin-Sommers and colleagues (2015) developed specific remediation approaches for different offender populations characterized by behavioral disinhibition (referred to as externalizers versus psychopaths). Indeed, they found that each group responded most strongly to the intervention targeting their purported underlying deficit (see also Baskin-Sommers & Newman, 2013). This finding highlights how, while individuals may share traits (e.g., disinhibition), those traits need not arise from equivalent mechanisms. As a result, individuals may undergo differential trajectories of change following treatment despite sharing traits on the behavioral level due to differences in underlying mechanisms contributing to those traits. To this end, incorporation of inclusion and exclusion criteria based on mechanistic indicators (whether cognitive or neurobiological, such as outlined in Perkins et al., 2020) in psychopathy research may be necessary to facilitate the development of effective remediation and treatment approaches. Additionally, analytic approaches such as component level analyses may be helpful for exploring the mechanisms of the broader Cleckleyan syndrome. Overall, with sufficient progress in specifying mechanisms, it should be possible to use mechanism-related correlates combined with symptom assessments like the PCL/-R (or trait assessments like the FFM) to specify psychopathy subtypes so that the treatment and prevention needs associated with the subtypes may be more efficiently addressed.

Future Directions

Use of various measures of psychopathy without attempts to reconcile conceptual or theoretical differences in assessments likely contributes to mixed findings regarding the structure of psychopathy, with results of null findings then interpreted as evidence against a given theory (e.g., Gunschera et al., 2023). It is worth noting that in their review of the neural correlates of psychopathy, Pujara et al. (2013) highlighted that integrating findings across studies and reconciling inconsistencies across studies is hindered by use of differing definitions of psychopathy, discrepant assessment methods, and inclusion of or focus on different subtypes. They also note that synthesis across studies is undermined by divergent analytical approaches based on conceptual understandings (i.e., group-based analyses based on the assumption that psychopathy relates to a unique mechanism of dysfunction versus dimensional analyses based on the assumption that psychopathy reflects variations in traits along graded continua). Since study results can differ as a function of approach (i.e., a college sample where psychopathy is being understood as relatively higher scores on a set of traits but where the rate of psychopathy as it would be understood in a clinical setting is very low), differences in the neuroimaging findings are difficult to interpret. We call for the iterative development and refinement of psychopathy assessment measures and theories in a way that ensures transparency of the construct being measured, whether that is Cleckleyan psychopathy or subcomponents of the psychopathy syndrome. We propose that such clarity and synthesis across different approaches will facilitate replicability and advancement while reducing null findings.

Research efforts must also endeavor to adopt a sociocultural perspective by exploring factors of individual and cultural diversity to better outline mechanisms and variations of the psychopathy syndrome. It is well-established that the PCL/-R performs differently cross-culturally (Cooke et al. 2005a), and there may indeed be ethnic and cultural variations in the manifestation of psychopathy (Cooke et al. 2005b; Fanti et al. 2018; Sullivan and Kosson 2006). Research demonstrates inconsistent findings in White/non-Hispanic male offenders compared to people of color or women. For example, while some studies show that high PCL/-R scores in female offenders are associated with deficits in emotion processing and aspects of attention (Anton et al., 2012; Maurer et al., 2016; Sutton et al., 2002; Vitale et al., 2007), female psychopathic samples have not reliably demonstrated predicted deficits in passive avoidance, response perseveration, and emotion lexical decision task performance (Efferson & Glenn, 2018; Maurer et al., 2016; Vitale et al., 2011; Vitale & Newman, 2001). Mixed findings are similarly apparent among African American samples (Baskin-Sommers et al. 2011a, b; Lorenz & Newman, 2002b). As a result, it has been difficult to determine if failures to support the RMH (and other causal models) in these samples is because (1) these mechanisms are not associated with psychopathy in these groups, (2) the PCL/-R was not capturing equivalent psychopathic individuals in these different groups, or (3) the expression of the mechanism/core underlying deficit is moderated by factors such as culture or socialization/the development of compensatory mechanisms (e.g., women socialized to be attentive to emotion). Inclusion of diverse samples would allow for cross-validation of measures in different samples and promote a cross-cultural understanding of psychopathy.

Summary & Conclusion

The PCL/-R was designed to capture the core interpersonal, affective, and behavioral attributes that exemplify the Cleckley psychopath and allow the systematic identification of a construct that had proven difficult to capture reliably. Using this instrument, researchers have been able to measure psychopathy reliably, observe clinically relevant correlates, demonstrate the discriminant validity of the construct, identify potential psychological and biological mechanisms for the disorder, formulate methods for assessing psychopathy risk in children, and propose treatment approaches that address specific abnormal processes in psychopathic subtypes. Accordingly, the PCL/-R has allowed considerable advancement in the field of psychopathy. As scientific growth has burgeoned with the use of the PCL/-R, other key developments in psychopathy involve the use of alternative measurement approaches. These alternative approaches have allowed researchers to identify a reliable method for predicting low-fear and other affect-related deficits and to characterize various components of psychopathy. While the greater diversity of research on psychopathy increases the tools and models available for assessing and conceptualizing psychopathy, it also presents researchers with the challenge of integrating diverse approaches and findings.

As already discussed, "psychopathy" is a heterogenous construct that may be defined differently by different measures and may be divided into subtypes. While these various conceptualizations of the syndrome will share overlapping traits and behaviors, other aspects of their construct validity are expected to differ. Thus, findings obtained using one method to assess psychopathy may have marginal or no relevance for findings obtained using another method. To help organize and integrate information pertaining to these diverse research strategies, it is crucial for researchers to specify the construct(s) being investigated and provide evidence for the construct validity of the specific construct(s) being studied (i.e., rather than assuming convergent and discriminant validity based on research with alternative constructs). We further argue that, considering the notion of equifinality (i.e., different pathways to a common outcome), the adoption of mechanism-focused approaches will allow for deeper understanding of the psychopathic syndrome. Through these approaches, the field can prevent a return to the confusion over the term 'psychopathy' that existed prior to the PCL/-R while also allowing for the development of alternative assessments of psychopathy.

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References

Acheson, S. K., & Payne, J. W. (2005). Review of the Psychopathy Checklist–Revised. In B. S. Plake, J. C. Impara, & R. A. Spies (Eds.), *Sixteenth mental measurements yearbook (Second* (pp. 429–431). Buros Institute of Mental Measurements.

Anderson, N. E., Steele, V. R., Maurer, J. M., Rao, V., Koenigs, M. R., Decety, J., & Kiehl, K. A. (2017). Differentiating emotional processing and attention in psychopathy with functional neuroimaging. *Cognitive Affective & Behavioral Neuroscience*, *17*, 491–515. https://doi.org/10.3758/s13415-016-0493-5

Anton, M. E., Baskin-Sommers, A. R., Vitale, J. E., Curtin, J. J., & Newman, J. P. (2012). Differential effects of psychopathy and antisocial personality disorder symptoms on cognitive and fear processing in female offenders. *Cognitive Affective and Behavioral Neuroscience*, *12*(4). https://doi.org/10.3758/s13415-012-0114-x

Arnett, P. A., Howland, E. W., Smith, S. S., & Newman, J. P. (1993). Autonomic responsivity during passive avoidance in incarcerated psychopaths. *Personality and Individual Differences*, *14*(1). https://doi.org/10.1016/0191-8869(93)90187-8

Baskin-Sommers, A. R., & Newman, J. P. (2013). Differentiating the cognition-emotion interactions that characterize psychopathy versus externalizing. *Handbook of cognition and emotion*, 501–520.

Baskin-Sommers, A. R., Curtin, J. J., & Newman, J. P. (2011a). Specifying the attentional selection that moderates the fearlessness of psychopathic offenders. *Psychological Science*, *22*(2). https://doi.org/10.1177/0956797610396227

Baskin-Sommers, A. R., Newman, J. P., Sathasivam, N., & Curtin, J. J. (2011b). Evaluating the generalizability of a fear deficit in psychopathic african american offenders. *Journal of Abnormal Psychology*, *120*(1). https://doi.org/10.1037/a0021225

Baskin-Sommers, A. R., Curtin, J. J., & Newman, J. P. (2013). Emotion-modulated startle in psychopathy: Clarifying familiar effects. *Journal of Abnormal Psychology*, *122*(2). https://doi.org/10.1037/a0030958

Baskin-Sommers, A. R., Curtin, J. J., & Newman, J. P. (2015). Altering the cognitive-affective dysfunctions of psychopathic and externalizing offender subtypes with cognitive remediation. *Clinical Psychological Science*, *3*(1). https://doi.org/10.1177/2167702614560744

Bechtel, W. (2009). Looking down, around, and up: Mechanistic explanation in psychology. *Philosophical Psychology*, *22*(5). https://doi.org/10.1080/09515080903238948

Benton, A. L., & Probst, K. A. (1946). A comparison of psychiatric ratings with Minnesota Multiphasic Personality Inventory scores. *Journal of Abnormal and Social Psychology*, *41*(1). https://doi.org/10.1037/h0060088

Bernstein, A., Newman, J. P., Wallace, J. F., & Luh, K. E. (2000). Left-hemisphere activation and deficient response modulation in psychopaths. *Psychological science*, *11*(5), 414–417. https://doi.org/10.1111/1467-9280.00280

Blackburn, R. (1983). Psychopathy, delinquency and crime. in individual differences and psychopathology. https://doi.org/10.1016/b978-0-12-273903-3.50017-1

Blackburn, R. (2009). Subtypes of psychopath. In Personality, personality disorder and violence: An evidence based approach.

Blair, R. J. R. (2009). The processing of emotional expression information in individuals with psychopathy. In Personality, personality disorder and violence: An evidence based approach.

Blair, R. J. R., Mitchell, D. G. V., Richell, R. A., Kelly, S., Leonard, A., Newman, C., & Scott, S. K. (2002). Turning a deaf ear to fear: Impaired recognition of vocal affect in psychopathic individuals. *Journal of Abnormal Psychology*, *111*(4), 682–686.

Braude, M. (1937). Constitutional psychopathy. *The principles and practice of clinical psychiatry* (pp. 238–254). P. Blakiston's Son & Co.

Brinkley, C. A., Newman, J. P., Widiger, T. A., & Lynam, D. R. (2004). Two approaches to parsing the heterogeneity of psychopathy. *Clinical Psychology: Science and Practice*, *11*(1). https://doi.org/10.1093/clipsy.bph054

Bryan, J. H., & Kapche, R. (1967). Psychopathy and verbal conditioning. *Journal of Abnormal Psychology*, *72*(1). https://doi.org/10.1037/h0024220

Choy, O., Raine, A., & Schug, R. (2022). Larger striatal volume is associated with increased adult psychopathy. *Journal of Psychiatric Research*. https://doi.org/10.1016/j.jpsychires.2022.03.006. 149.

Cleckley, H. (1941). *The Mask of Sanity*. Mosby.

Cleckley, H. (1988). *The Mask of Sanity (Fifth)*. Emily S. Cleckley.

Contreras-Rodríguez, O., Pujol, J., Batalla, I., Harrison, B. J., Soriano-Mas, C., Deus, J., López-Solà, M., Macià, D., Pera, V., Hernández-Ribas, R., Pifarré, J., Menchón, J. M., & Cardoner, N. (2015). Functional connectivity bias in the prefrontal cortex of psychopaths. *Biological Psychiatry*, *78*(9). https://doi.org/10.1016/j.biopsych.2014.03.007

Cooke, D. J., & Michie, C. (1997). An item response theory analysis of the hare psychopathy checklist - Revised. *Psychological Assessment*, *9*(1). https://doi.org/10.1037/1040-3590.9.1.3

Cooke, D. J., Michie, C., Hart, S. D., & Clark, D. (2005a). Assessing psychopathy in the UK: Concerns about cross-cultural generalisability. British Journal of Psychiatry, 186(APR.). https://doi.org/10.1192/bjp.186.4.335

Cooke, D. J., Michie, C., Hart, S. D., & Clark, D. (2005b). Searching for the pan-cultural core of psychopathic personality disorder. *Personality and Individual Differences*, *39*(2). https://doi.org/10.1016/j.paid.2005.01.004

Craig, M. C., Catani, M., Deeley, Q., Latham, R., Daly, E., Kanaan, R., & Murphy, D. G. (2009). Altered connections on the road to psychopathy. *Molecular psychiatry*, *14*(10), 946–953.

Craver, C. (2007). *Explaining the Brain*. Oxford University Press.

Craver, C., & Bechtel, W. (2006). Mechanism. In S. Sarkar, & J. Pfeifer (Eds.), *Philosophy of Science: An Encyclopedia* (pp. 469–478). Routledge.

Craver, C., & Tabery, J. (2023). Mechanisms in Science. In E. N. Zalta & U. Nodelman (Eds.), The Stanford Encyclopedia of Philosophy. https://plato.stanford.edu/entries/science-mechanisms/

de Vogel, V., & Lancel, M. (2016). Gender differences in the assessment and manifestation of psychopathy: Results from a multicenter study in forensic psychiatric patients. *International Journal of Forensic Mental Health*, *15*(1). https://doi.org/10.1080/14999013.2016.1138173

Decety, J., Skelly, L., Yoder, K. J., & Kiehl, K. A. (2014). Neural processing of dynamic emotional facial expressions in psychopaths. *Social neuroscience*, *9*(1), 36–49.

DeMatteo, D., & Edens, J. F. (2006). The role and relevance of the psychopathy checklist-revised in court: A case law survey of U.S. courts (1991–2004). *Psychology Public Policy and Law*, *12*(2). https://doi.org/10.1037/1076-8971.12.2.214

Deming, P., Cook, C. J., Meyerand, M. E., Kiehl, K. A., Kosson, D. S., & Koenigs, M. (2023). Impaired salience network switching in

psychopathy. *Behavioural Brain Research*, 452. https://doi.org/10.1016/j.bbr.2023.114570

Derefinko, K., & Lynam, D. R. (2013). Psychopathy from the perspective of the five-factor model of personality. In T. A. Widiger, & P. T. Costa Jr. (Eds.), *Personality disorders and the five-factor model of personality* (3rd ed., pp. 103–117). American Psychological Association. https://doi.org/10.1037/13939-007

Doctor, R. M., & Craine, W. H. (1971). Modification of drug language usage of primary and neurotic psychopaths. *Journal of Abnormal Psychology*, *77*(2). https://doi.org/10.1037/h0030733

Dolan, M., & Doyle, M. (2000). Violence risk prediction: Clinical and actuarial measures and the role of the Psychopathy Checklist. *British Journal of Psychiatry*, *177*. https://doi.org/10.1192/bjp.177.4.303

Driver, J. (2001). A selective review of selective attention research from the past century. *British Journal of Psychology*, *92*(1). https://doi.org/10.1348/000712601162103

Dvorak-Bertsch, J. D., Sadeh, N., Glass, S. J., Thornton, D., & Newman, J. P. (2007). Stroop tasks associated with differential activation of anterior cingulate do not differentiate psychopathic and non-psychopathic offenders. *Personality and individual differences*, *42*(3), 585–595. https://doi.org/10.1016/j.paid.2006.07.023

Edwards, B. G., Gullapalli, A. R., Maurer, J. M., Ulrich, D. M., Harenski, C. L., Thomson, N. D., Davenport, S., & Kiehl, K. A. (2023). Psychopathy and substance use predict recidivism in women: A 7-year prospective Study. *Personality Disorders: Theory Research and Treatment*, *14*(4). https://doi.org/10.1037/per0000615

Efferson, L. M., & Glenn, A. L. (2018). Examining gender differences in the correlates of psychopathy: A systematic review of emotional, cognitive, and morality-related constructs. *Aggression and Violent Behavior*, *41*. https://doi.org/10.1016/j.avb.2018.05.009

Espinoza, F. A., Anderson, N. E., Vergara, V. M., Harenski, C. L., Decety, J., Rachakonda, S., Damaraju, E., Koenigs, M., Kosson, D. S., Harenski, K., Calhoun, V. D., & Kiehl, K. A. (2019). *Resting-state fMRI dynamic functional network connectivity and associations with psychopathy traits* (p. 24). Clinical. https://doi.org/10.1016/j.nicl.2019.101970

Fanti, K. A., Lordos, A., Sullivan, E. A., & Kosson, D. S. (2018). Cultural and ethnic variations in psychopathy. Handbook of Psychopathy., 2nd Ed.

Gernsbacher, M. A., & Faust, M. (1991). The role of suppression in sentence comprehension. *Advances in psychology*, *77*, 97–128.

Gilbert, G. M. (1945). A preview of veteran adjustment problems. *Journal of Abnormal and Social Psychology*, *40*(3). https://doi.org/10.1037/h0056102

Glennan, S. (2005). Modeling mechanisms. Studies in history and philosophy of science Part C: studies in history and philosophy of. *Biological and Biomedical Sciences*, *36*(2). https://doi.org/10.1016/j.shpsc.2005.03.011

Gorenstein, E. E., & Newman, J. P. (1980). Disinhibitory psychopathology: A new perspective and a model for research. *Psychological Review*, *87*(3). https://doi.org/10.1037/0033-295X.87.3.301

Gorman, B. S. (1970). 16 PF correlates of sensation-seeking. *Psychological Reports*, *26*(3). https://doi.org/10.2466/pr0.1970.26.3.741

Grahek, I., Schaller, M., & Tackett, J. L. (2021). Anatomy of a psychological theory: integrating construct-validation and computational-modeling methods to advance theorizing. *Perspectives on Psychological Science*, *16*(4). https://doi.org/10.1177/1745691620966794

Gunschera, L. J., Verschuere, B., Murphy, R. A., Temple-McCune, A., Dutton, K., & Fox, E. (2023). No impaired integration in psychopathy: Evidence from an illusory conjunction paradigm. *Personality Disorders: Theory Research and Treatment*, *14*(5). https://doi.org/10.1037/per0000619

Hamilton, R. K. B., & Newman, J. P. (2018). Information processing capacity in psychopathy: Effects of anomalous attention. *Personality Disorders: Theory Research and Treatment*, *9*(2). https://doi.org/10.1037/per0000223

Hamilton, R. K. B., Baskin-Sommers, A. R., & Newman, J. P. (2014). Relation of frontal N100 to psychopathy-related differences in selective attention. *Biological Psychology*, 103. https://doi.org/10.1016/j.biopsycho.2014.08.012

Hamilton, R. K. B., Racer, K. H., & Newman, J. P. (2015). Impaired integration in psychopathy: A unified theory of psychopathic dysfunction. *Psychological Review*, *122*(4). https://doi.org/10.1037/a0039703

Hare, R. D. (1972). Psychopathy and physiological responses to adrenalin. *Journal of Abnormal Psychology*, *79*(2). https://doi.org/10.1037/h0032725

Hare, R. D. (1978). Psychopathy and electrodermal responses to nonsignal stimulation. *Biological Psychology*, *6*(4). https://doi.org/10.1016/0301-0511(78)90026-1

Hare, R. D. (1979). Psychopathy and laterality of cerebral function. *Journal of Abnormal Psychology*, *88*(6). https://doi.org/10.1037/0021-843X.88.6.605

Hare, R. D. (1980). A research scale for the assessment of psychopathy in criminal populations. *Personality and Individual Differences*, *1*(2), 111–119. https://doi.org/10.1016/0191-8869(80)90028-8

Hare, R. D. (1991a). *L'échelle de psychopathie de Hare-révisée (PCL-R)*. Multi-Health Systems.

Hare, R. D. (1991b). *Manual for the Revised Psychopathy Checklist* (1st ed.). Multi-Health Systems.

Hare, R. D. (1996). Psychopathy: A clinical construct whose time has come. *Criminal Justice and Behavior*, *23*(1). https://doi.org/10.1177/0093854896023001004

Hare, R. D. (1998). Psychopaths and their nature: Implications for the mental health and criminal justice systems. In T. Millon, E. Simonson, M. Burket-Smith, & R. Davis (Eds.), *Psychopathy: Antisocial, criminal, and violent behavior* (pp. 188–212). Guilford Press.

Hare, R. D. (2003). Manual for the Revised Psychopathy Checklist (Second Edition). Multi-Health Systems.

Hare, R. D., & Neumann, C. S. (2008). Psychopathy as a clinical and empirical construct. *Annual Review of Clinical Psychology*. https://doi.org/10.1146/annurev.clinpsy.3.022806.091452. 4.

Hare, R. D., & Neumann, C. S. (2010). The role of antisociality in the psychopathy construct: Comment on skeem and cooke (2010). *Psychological Assessment*, *22*(2). https://doi.org/10.1037/a0013635

Hare, R. D., Frazelle, J., & Cox, D. N. (1978). Psychopathy and physiological responses to threat of an aversive stimulus. *Psychophysiology*, *15*(2). https://doi.org/10.1111/j.1469-8986.1978.tb01356.x

Hare, R. D., Harpur, T. J., Hakstian, A. R., Forth, A. E., Hart, S. D., & Newman, J. P. (1990). The Revised Psychopathy Checklist: Reliability and Factor Structure. *Psychological Assessment*, *2*(3). https://doi.org/10.1037/1040-3590.2.3.338

Hare, R. D., Clark, D., Grann, M., & Thornton, D. (2000). Psychopathy and the predictive validity of the PCL-R: An international perspective. *Behavioral Sciences and the Law*, *18*(5). https://doi.org/10.1002/1099-0798(200010)18:5%3C623::AID-BSL409%3E3.0.CO;2-W

Hare, R. D., Neumann, C. S., & Mokros, A. (2018). The PCL-R assessment of psychopathy: Development, properties, debates, and new directions. In C. J. Patrick (Ed.), *Handbook of psychopathy* (2nd edn., pp. 39–79). Guilford Press.

Hauser, N. C., Neumann, C. S., Marshall, J., & Mokros, A. (2021). Rational, emotional, or both? Subcomponents of psychopathy predict opposing moral decisions. *Behavioral Sciences and the Law*, *39*(5). https://doi.org/10.1002/bsl.2547

Healy, W. (1942). Review of The mask of sanity, an attempt to reinterpret the so-called psychopathic personality. *The Journal of Abnormal and Social Psychology*, *37*(1). https://doi.org/10.1037/h0053042

Heilbrun, A. B. (1979). Psychopathy and violent crime. *Journal of Consulting and Clinical Psychology*, *47*(3). https://doi.org/10.1037/0022-006X.47.3.509

Hemphill, J. F., Hare, R. D., & Wong, S. (1998). Psychopathy and recidivism: A review. *Legal and criminological Psychology*, *3*(1), 139–170. https://doi.org/10.1111/j.2044-8333.1998.tb00355.x

Henderson, D. K. (1947). *Psychopathic states (Second Edition)*. W.W. Norton & Company.

Hervé, H. (2007). Psychopathic subtypes: Historical and contemporary perspectives. In H. Hervé, & J. C. Yuille (Eds.), *The psychopath: Theory, research, and practice* (pp. 431–460). Erlbaum.

Hervé, H. (2017). Psychopathy across the ages: A history of the Hare psychopath. In *The psychopath* (pp. 31–55). Routledge. https://doi.org/10.4324/9781315085470-2

Hetherington, E. M., & Klinger, E. (1964). Psychopathy and punishment. *Journal of Abnormal and Social Psychology*, *69*(1). https://doi.org/10.1037/h0039798

Hiatt, K. D., Schmitt, W. A., & Newman, J. P. (2004). Stroop tasks reveal abnormal selective attention among psychopathic offenders. *Neuropsychology*, *18*(1). https://doi.org/10.1037/0894-4105.18.1.50

Hicks, B. M., & Drislane, L. E. (2018). Variants (subtypes) of psychopathy. In C. J. Patrick (Ed.), *Handbook of psychopathy* (2nd edn., pp. 297–334). Guilford Press.

Hofmann, M. J., Mokros, A., & Schneider, S. (2024). The joy of being frightened: Fear experience in psychopathy. *Journal of Personality*, *92*(2). https://doi.org/10.1111/jopy.12890

House, T. H., & Milligan, W. L. (1976). Autonomic responses to modeled distress in prison psychopaths. *Journal of Personality and Social Psychology*, *34*(4). https://doi.org/10.1037/0022-3514.34.4.556

Howland, E. W., Kosson, D. S., Patterson, C. M., & Newman, J. P. (1993). Altering a Dominant Response: Performance of Psychopaths and Low-Socialization College Students on a Cued Reaction Time Task. *Journal of Abnormal Psychology*, *102*(3). https://doi.org/10.1037/0021-843X.102.3.379

Hoyle, R. H., Lynam, D. R., Miller, J. D., & Pek, J. (2023). The questionable practice of partialing to refine scores on and inferences about measures of psychological constructs. *Annual Review of Clinical Psychology*, *19*, 155–176. https://doi.org/10.1146/annurev-clinpsy-071720-015436

Hundleby, J. D., & Ross, B. E. (1977). Comparison of measures of psychopathy. *Journal of Consulting and Clinical Psychology*, *45*(4). https://doi.org/10.1037/0022-006X.45.4.702

Jakobwitz, S., & Egan, V. (2006). The dark triad and normal personality traits. *Personality and Individual Differences*, *40*(2). https://doi.org/10.1016/j.paid.2005.07.006

Karpman, B. (1941). On the need of separating psychopathy into two distinct clinical types: the symptomatic and the idiopathic. *Journal of Criminal Psychopathology*, *3*, 112–137.

Karpman, B. (1948). The myth of the psychopathic personality. *The American Journal of Psychiatry*, *104*(9). https://doi.org/10.1176/ajp.104.9.523

Kennealy, P. J., Hicks, B. M., & Patrick, C. J. (2007). Validity of factors of the Psychopathy Checklist—Revised in female prisoners: Discriminant relations with antisocial behavior, substance abuse, and personality. *Assessment*, *14*(4), 323–340. https://doi.org/10.1177/1073191107305882

Koenigs, M., Baskin-Sommers, A., Zeier, J., & Newman, J. P. (2011). Investigating the neural correlates of psychopathy: A critical review. *Molecular Psychiatry*, *16*(8). https://doi.org/10.1038/mp.2010.124

Korponay, C., & Koenigs, M. (2021). Gray matter correlates of impulsivity in psychopathy and in the general population differ by kind, not by degree: A comparison of systematic reviews. *Social Cognitive and Affective Neuroscience*, *16*(7). https://doi.org/10.1093/scan/nsab045

Kosson, D. S. (1996). Psychopathy and dual-task performance under focusing conditions. *Journal of Abnormal Psychology*, *105*(3). https://doi.org/10.1037/0021-843X.105.3.391

Kosson, D. S. (1998). Divided visual attention in psychopathic and nonpsychopathic offenders. *Personality and Individual Differences*, *24*(3). https://doi.org/10.1016/S0191-8869(97)00176-1

Kosson, D. S., Miller, S. K., Byrnes, K. A., & Leveroni, C. L. (2007). Testing neuropsychological hypotheses for cognitive deficits in psychopathic criminals: A study of global-local processing. *Journal of the International Neuropsychological Society*, *13*(2). https://doi.org/10.1017/S1355617707070294

Krusemark, E. A., Kiehl, K. A., & Newman, J. P. (2016). Endogenous attention modulates early selective attention in psychopathy: An ERP investigation. *Cognitive Affective and Behavioral Neuroscience*, *16*(5). https://doi.org/10.3758/s13415-016-0430-7

Lake, A. J., Baskin-Sommers, A. R., Li, W., Curtin, J. J., & Newman, J. P. (2011). Evidence for unique threat-processing mechanisms in psychopathic and anxious individuals. *Cognitive Affective and Behavioral Neuroscience*, *11*(4). https://doi.org/10.3758/s13415-011-0041-2

Larson, C. L., Baskin-Sommers, A. R., Stout, D. M., Balderston, N. L., Curtin, J. J., Schultz, D. H., Kiehl, K. A., & Newman, J. P. (2013). The interplay of attention and emotion: Top-down attention modulates amygdala activation in psychopathy. *Cognitive Affective and Behavioral Neuroscience*, *13*(4). https://doi.org/10.3758/s13415-013-0172-8

Le, N. N. H., Miller, H. A., & Marshall, E. A. (2023). Examining psychopathy as a predictor of recidivism in a sample of women who have sexually offended. *Journal of Criminal Justice*, 88. https://doi.org/10.1016/j.jcrimjus.2023.102094

Lilienfeld, S. O. (2018). The multidimensional nature of psychopathy: Five recommendations for research. *Journal of Psychopathology and Behavioral Assessment*, *40*(1). https://doi.org/10.1007/s10862-018-9657-7

Lilienfeld, S. O., Watts, A. L., Francis Smith, S., Berg, J. M., & Latzman, R. D. (2015). Psychopathy Deconstructed and Reconstructed: Identifying and Assembling the Personality Building Blocks of Cleckley's Chimera. *Journal of Personality*, *83*(6). https://doi.org/10.1111/jopy.12118

Lindner, P., Flodin, P., Budhiraja, M., Savic, I., Jokinen, J., Tiihonen, J., & Hodgins, S. (2018). Associations of psychopathic traits with local and global brain network topology in young adult women. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, *3*(12). https://doi.org/10.1016/j.bpsc.2018.04.010

Llanes, S. J., & Kosson, D. S. (2006). Divided visual attention and left hemisphere activation among psychopathic and nonpsychopathic offenders. *Journal of Psychopathology and Behavioral Assessment*, *28*(1). https://doi.org/10.1007/s10862-006-4533-2

Lorenz, A. R., & Newman, J. P. (2002a). Deficient response modulation and emotion processing in low-anxious Caucasian psychopathic offenders: Results from a lexical decision task. *Emotion*, *2*(2). https://doi.org/10.1037/1528-3542.2.2.91

Lorenz, A. R., & Newman, J. P. (2002b). Do emotion and information processing deficiencies found in Caucasian psychopaths generalize to African-American psychopaths? *Personality and Individual Differences*, *32*(6). https://doi.org/10.1016/S0191-8869(01)00111-8

Lowrey, L. G. (1943). Review of Mask of Sanity. *American Journal of Orthopsychiatry*, *13*(2), 374–374. https://doi.org/10.1111/j.1939-0025.1943.tb06008.x

Lykken, D. T. (1957). A study of anxiety in the sociopathic personality. *Journal of Abnormal and Social Psychology*, *55*(1). https://doi.org/10.1037/h0047232

Lykken, D. T. (1995). The Antisocial Personalities (First Edition). ImprintPsychology Press. https://doi.org/10.4324/9780203763551

Lykken, D. T. (2006). Psychopathic Personality: The Scope of the Problem. In C. J. Patrick (Ed.), *Handbook of psychopathy* (pp. 3–13). The Guilford Press.

Lynam, D. R., & Miller, J. D. (2015). Psychopathy from a basic trait perspective: the utility of a five-factor model approach. *Journal of Personality*, *83*(6). https://doi.org/10.1111/jopy.12132

Lynam, D. R., & Widiger, T. A. (2007). Using a general model of personality to identify the basic elements of psychopathy. *Journal of Personality Disorders*, *21*(2). https://doi.org/10.1521/pedi.2007.21.2.160

Lynam, D. R., Hoyle, R. H., & Newman, J. P. (2006). The perils of partialling: Cautionary tales from aggression and psychopathy. *Assessment*, *13*(3). https://doi.org/10.1177/1073191106290562

Macdonald, J. M. (1966). Psychopaths Unlimited. Contemporary Psychology: A Journal of Reviews, 11(I2), 88–89. https://resolver.scholarsportal.info/resolve/00107549/v11i0002/88_pu.xml

Maher, B. (1973). *Contemporary abnormal psychology*. Penguin Books Ltd.

Maurer, M., Steele, V. R., Edwards, B. G., Bernat, E. M., Calhoun, V. D., & Kiehl, K. A. (2016). Dysfunctional error-related processing in female psychopathy. *Social Cognitive and Affective Neuroscience*, *11*(7). https://doi.org/10.1093/scan/nsv070

McCarthy, A., Fox, B., & Verona, E. (2024). The relationship between psychopathy facets and types of criminal offences. *Journal of Investigative Psychology and Offender Profiling*, *21*(2), Article e1628, https://doi.org/10.1002/jip.1628.

McCord, W., & McCord, J. (1964). *The psychopath: An essay on the criminal mind*. D. Van Nostrand.

Miller, J. D., & Lynam, D. R. (2015). Understanding psychopathy using the basic elements of personality. *Social and Personality Psychology Compass*, *9*(5), 223–237. https://doi.org/10.1111/spc3.12170

Miller, J. D., Wilson, L. F., Hyatt, C. S., & Zeichner, A. (2015). Psychopathic traits and aggression: Which trait components predict aggressive responding in a laboratory task? *Personality and Individual Differences*, 87. https://doi.org/10.1016/j.paid.2015.08.008

Moosburner, M., Etzler, S., Brunner, F., Briken, P., & Rettenberger, M. (2024). Is psychopathy a dynamic risk factor? an empirical investigation of changes in psychopathic personality traits over the course of correctional treatment. *Criminal Justice and Behavior*, *51*(2). https://doi.org/10.1177/00938548231219804

Moretti, G., Flutti, E., Colanino, M., Ferlito, D., Amoresano, L., & Travaini, G. (2024). Recidivism risk in male adult sex offenders with psychopathic traits assessed by PCL-R: A systematic review. *Medicine Science and the Law*, *64*(1), 41–51. https://doi.org/10.1177/00258024231187186

Morey, L. C. (1991). Classification of mental disorder as a collection of hypothetical constructs. *Journal of Abnormal Psychology*, *100*(3). https://doi.org/10.1037/0021-843X.100.3.289

Newman, J. P. (1998). Psychopathic behavior: An information processing perspective. *Psychopathy: Theory, research and implications for society* (pp. 81–104). Springer Netherlands.

Newman, J. P., & Kosson, D. S. (1986). Passive avoidance learning in psychopathic and nonpsychopathic offenders. *Journal of Abnormal Psychology*, *95*(3). https://doi.org/10.1037/0021-843X.95.3.252

Newman, J. P., Gorenstein, E. E., & Kelsey, J. E. (1983). Failure to delay gratification following septal lesions in rats: Implications for an animal model of disinhibitory psychopathology. *Personality and Individual Differences*, *4*(2). https://doi.org/10.1016/0191-8869(83)90014-4

Newman, J. P., Widom, C. S., & Nathan, S. (1985). Passive avoidance in syndromes of disinhibition: Psychopathy and extraversion. *Journal of Personality and Social Psychology*, *48*(5). https://doi.org/10.1037//0022-3514.48.5.1316

Newman, J. P., Patterson, C. M., & Kosson, D. S. (1987). Response Perseveration in Psychopaths. *Journal of Abnormal Psychology*, *96*(2). https://doi.org/10.1037/0021-843X.96.2.145

Newman, J. P., Patterson, C. M., Howland, E. W., & Nichols, S. L. (1990). Passive avoidance in psychopaths: The effects of reward. *Personality and Individual Differences*, *11*(11). https://doi.org/10.1016/0191-8869(90)90021-I

Newman, J. P., Schmitt, W. A., & Voss, W. D. (1997). The impact of motivationally neutral cues on psychopathic individuals: Assessing the generality of the response modulation hypothesis. *Journal of Abnormal Psychology*, *106*(4). https://doi.org/10.1037/0021-843X.106.4.563

Newman, J. P., MacCoon, D. G., Vaughn, L. J., & Sadeh, N. (2005). Validating a distinction between primary and secondary psychopathy with measures of Gray's BIS and BAS constructs. *Journal of Abnormal Psychology*, *114*(2). https://doi.org/10.1037/0021-843X.114.2.319

Patrick, C. J. (2017). Getting to the Heart of Psychopathy. *The Psychopath: Theory, Research, and Practice* (pp. 207–252). Routledge Taylor & Francis Group, LLC.

Patrick, C. J. (2022). Psychopathy: Current knowledge and future directions. *Annual Review of Clinical Psychology*, *18*, 387–415. https://doi.org/10.1146/annurev-clinpsy-072720-012851

Patrick, C., Drislane, L. E., & Strickland, C. (2012). Conceptualizing psychopathy in triarchic terms: Implications for treatment. *International Journal of Forensic Mental Health*, *11*(4). https://doi.org/10.1080/14999013.2012.746761

Patterson, C. M., & Newman, J. P. (1993). Reflectivity and learning from aversive events: Toward a psychological mechanism for the syndromes of disinhibition. *Psychological Review*, *100*(4). https://doi.org/10.1037/0033-295x.100.4.716

Perkins, E. R., Latzman, R. D., & Patrick, C. J. (2020). Interfacing neural constructs with the hierarchical taxonomy of psychopathology: 'Why" and how"'. *Personality and Mental Health*, *14*(1). https://doi.org/10.1002/pmh.1460

Philippi, C. L., Pujara, M. S., Motzkin, J. C., Newman, J., Kiehl, K. A., & Koenigs, M. (2015). Altered resting-state functional connectivity in cortical networks in psychopathy. *Journal of Neuroscience*, *35*(15). https://doi.org/10.1523/JNEUROSCI.5010-14.2015

Porter, S. (1996). Without conscience or without active conscience? The etiology of psychopathy revisited. *Aggression and Violent Behavior*, *1*(2). https://doi.org/10.1016/1359-1789(95)00010-0

Pujara, M., Motzkin, J. C., Newman, J. P., Kiehl, K. A., & Koenigs, M. (2013). Neural correlates of reward and loss sensitivity in psychopathy. *Social Cognitive and Affective Neuroscience*, *9*(6). https://doi.org/10.1093/scan/nst054

Raskin, D. C., & Hare, R. D. (1978). Psychopathy and detection of deception in a prison population. *Psychophysiology*, *15*(2). https://doi.org/10.1111/j.1469-8986.1978.tb01348.x

Rimé, B., Bouvy, H., Leborgne, B., & Rouillon, F. (1978). Psychopathy and nonverbal behavior in an interpersonal situation. *Journal of Abnormal Psychology*, *87*(6). https://doi.org/10.1037/0021-843X.87.6.636

Rotenberg, M. (1975). Psychopathy, insensitivity and sensitization. *Professional Psychology: Research and Practice*, *6*(3). https://doi.org/10.1037/0735-7028.6.3.283

Rubin, H. (1948). The Minnesota Multiphasic Personality Inventory as a diagnostic aid in a veterans hospital. *Journal of Consulting Psychology*, *12*(4). https://doi.org/10.1037/h0055299

Sandvik, A. M., Hansen, A. L., Johnsen, B. H., & Laberg, J. C. (2014). Psychopathy and the ability to read the language of the eyes: Divergence in the psychopathy construct. *Scandinavian Journal of Psychology*, *55*(6). https://doi.org/10.1111/sjop.12138

Schachter, S., & Latané, B. (1964). Crime, cognition and the autonomic nervous system. In M. R. Jones (Ed.), Nebraska Symposium on Motivation. University of Nebraska Press.

Schalling, D., & Rosén, A. S. S. (1968). Porteus Maze Differences Between Psychopathic and Non-psychopathic Criminals. *British Journal of Social and Clinical Psychology*, *7*(3). https://doi.org/10.1111/j.2044-8260.1968.tb00562.x

Schmitt, W. A., & Newman, J. P. (1999). Are all psychopathic individuals low-anxious? *Journal of Abnormal Psychology*, *108*(2). https://doi.org/10.1037/0021-843X.108.2.353

Sheng, T., Gheytanchi, A., & Aziz-Zadeh, L. (2010). Default network deactivations are correlated with psychopathic personality traits. *Plos One*, *5*(9). https://doi.org/10.1371/journal.pone.0012611

Skeem, J. L., & Cooke, D. J. (2010). Is criminal behavior a central component of psychopathy? conceptual directions for resolving the Debate. *Psychological Assessment*, *22*(2). https://doi.org/10.1037/a0008512

Smith, S. S., & Newman, J. P. (1990). Alcohol and drug abuse-dependence disorders in psychopathic and nonpsychopathic criminal offenders. *Journal of Abnormal Psychology*, *99*(4). https://doi.org/10.1037/0021-843X.99.4.430

Smith, G. T., McCarthy, D. M., & Zapolski, T. C. B. (2009). On the Value of Homogeneous Constructs for Construct Validation, Theory Testing, and the Description of Psychopathology. *Psychological Assessment*, *21*(3). https://doi.org/10.1037/a0016699

Sohn, J. S., & Lee, S. J. (2023). Revisiting the Relationships between Psychopathy Checklist-Revised (PCL-R) Facets and Sexual Recidivists versus Nonsexual Recidivists. *Journal of Forensic Psychology Research and Practice*, *23*(4). https://doi.org/10.1080/24732850.2022.2044712

Sohn, J. S., Raine, A., & Hong, Y. O. (2022). A Link between Psychopathy Affect and Instrumentality in Homicide. *Homicide Studies*, *26*(3). https://doi.org/10.1177/10887679211028879

Steinberg, E. P., & Schwartz, G. E. (1976). Biofeedback and electrodermal self-regulation in psychopathy. *Journal of Abnormal Psychology*, *85*(4). https://doi.org/10.1037/0021-843X.85.4.408

Strickland, C. M., Drislane, L. E., Lucy, M., Krueger, R. F., & Patrick, C. J. (2013). Characterizing Psychopathy Using DSM-5 Personality Traits. *Assessment*, *20*(3). https://doi.org/10.1177/1073191113486691

Sullivan, E. A., & Kosson, D. S. (2006). Ethnic and Cultural Variations in Psychopathy. In C. J. Patrick (Ed.), *Handbook of Psychopathy* (pp. 437–458). The Guilford Press.

Sumner, F. C. (1941). Review of The mask of sanity. *Psychological Bulletin*, *38*(9), 904–905.

Sutton, S. K., Vitale, J. E., & Newman, J. P. (2002). Emotion among women with psychopathy during picture perception. *Journal of Abnormal Psychology*, *111*(4). https://doi.org/10.1037/0021-843X.111.4.610

Thagard, P. (1998). Explaining Disease: Correlations, Causes, and Mechanisms. *Minds and Machines*, *8*(1). https://doi.org/10.1023/A:1008286314688

Thorne, F. C. (1947). Etiological studies of psychopathic personality: the ego-inflated, defectively-conditioned type. *Journal of Consulting Psychology*, *11*(6). https://doi.org/10.1037/h0061095

Tillem, S., van Dongen, J., Brazil, I. A., & Baskin-Sommers, A. (2018). Psychopathic traits are differentially associated with efficiency of neural communication. *Psychophysiology*, *55*(9). https://doi.org/10.1111/psyp.13194

Vitale, J. E., & Newman, J. P. (2001). Response perseveration in psychopathic women. *Journal of Abnormal Psychology*, *110*(4). https://doi.org/10.1037/0021-843X.110.4.644

Vitale, J. E., Brinkley, C. A., Hiatt, K. D., & Newman, J. P. (2007). Abnormal selective attention in psychopathic female offenders. *Neuropsychology*, *21*(3). https://doi.org/10.1037/0894-4105.21.3.301

Vitale, J. E., Maccoon, D. G., & Newman, J. P. (2011). Emotion facilitation and passive avoidance learning in psychopathic female offenders. *Criminal Justice and Behavior*, *38*(7). https://doi.org/10.1177/0093854811403590

Wallace, J. F., Vitale, J. E., & Newman, J. P. (1999). Response modulation deficits: Implications for the diagnosis and treatment of psychopathy. *Journal of Cognitive Psychotherapy*, *13*(1), 55–70. https://doi.org/10.1891/0889-8391.13.1.55

Whitelock, P. R., Overall, J. E., & Patrick, J. H. (1971). Personality patterns and alcohol abuse in a state hospital population. *Journal of Abnormal Psychology*, *78*(1). https://doi.org/10.1037/h0031483

Whitely, S. E. (1983). Construct validity: Construct representation versus nomothetic span. *Psychological Bulletin*, *93*(1). https://doi.org/10.1037/0033-2909.93.1.179

Widiger, T. A., & Lynam, D. R. (1998). Psychopathy as a variant of common personality traits: Implications for diagnosis, etiology, and pathology. In T. Millon (Ed.), *Psychopathy: Antisocial, criminal, and violent behavior* (pp. 171–187). Guilford.

Widom, C. S. (1976). Interpersonal and personal construct systems in psychopaths. *Journal of Consulting and Clinical Psychology*, *44*(4). https://doi.org/10.1037/0022-006X.44.4.614

Wilson, J. G., & Persor, M. J. (1939). The psychopathic prisoner. *Problems in prison psychiatry* (pp. 122–143). Caxton Printers.

Wolf, R. C., Carpenter, R. W., Warren, C. M., Zeier, J. D., Baskin-Sommers, A. R., & Newman, J. P. (2012). Reduced Susceptibility to the Attentional Blink in Psychopathic Offenders: Implications for the Attention Bottleneck Hypothesis. *Neuropsychology*, *26*(1). https://doi.org/10.1037/a0026000

Yang, Y., Raine, A., Joshi, A. A., Joshi, S., Chang, Y. T., Schug, R. A., Wheland, D., Leahy, R., & Narr, K. L. (2012). Frontal information flow and connectivity in psychopathy. *British Journal of Psychiatry*, *201*(5). https://doi.org/10.1192/bjp.bp.111.107128

Yoder, K. J., Harenski, C. L., Kiehl, K. A., & Decety, J. (2022). *Psychopathic traits modulate functional connectivity during pain perception and perspective-taking in female inmates* (p. 34). Clinical. https://doi.org/10.1016/j.nicl.2022.102984

Zeier, J. D., & Newman, J. P. (2013). Feature-based attention and conflict monitoring in criminal offenders: Interactive relations of psychopathy with anxiety and externalizing. *Journal of Abnormal Psychology*, *122*(3). https://doi.org/10.1037/a0033873

Zeier, J. D., Maxwell, J. S., & Newman, J. P. (2009). Attention Moderates the Processing of Inhibitory Information in Primary Psychopathy. *Journal of Abnormal Psychology*, *118*(3). https://doi.org/10.1037/a0016480

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